

for the Kempner function

Self-contained theorem, proof, sharpness construction, consequence for the Tuřescu conjecture, formal verification record, computational audit, and bibliographic novelty assessment.

PROVED	Sharp coprime-fiber inequality, local cross-exponent bounds, and infinite sharp family.
FORMALIZED	Lean/Mathlib checks the actual Kempner graph predicate and the main global theorem.
OPEN	The full Tuřescu conjecture $S(n) \neq S(n+1)$ is not resolved.
NOVELTY	Apparently new after an expanded database audit; historical novelty is not formally certifiable.

One-sentence handoff

Main result

If two nontrivial coprime integers x and y have the same Kempner value m , then the common value is sharply constrained by the largest prime exponents in x and y : $m + \min(E(x), E(y)) \leq E(x)E(y)$. This forces every hypothetical consecutive-value counterexample into a square-root-scale smooth-number set.

What this note does not claim

- It does not prove the full conjecture $S(n) \neq S(n+1)$.
- Lean verification certifies mathematical derivations relative to Lean/Mathlib; it does not certify bibliographic novelty.
- The computation validates the theorem and searches a large finite range; it is not used in the proof.

Audit date: 2026-08-20 | Workspace artifact: KempnerResearch

1. Definitions and theorem

For an integer $n \geq 1$, the Kempner function (often called the Smarandache function in the later literature) is

$$S(n) = \min \{ m \geq 1 : n \text{ divides } m! \}$$

For a prime p , let $v_p(z)$ be the exponent of p in z . For $z > 1$ define

$$E(z) = \max \{ v_l(z) : l \text{ is prime and } l \text{ divides } z \}$$

$$P_{\text{plus}}(z) = \text{largest prime factor of } z$$

Global coprime-fiber theorem

Theorem 1

Let $x, y > 1$ be coprime and suppose $S(x)=S(y)=m$. Then

$$m + \min(E(x), E(y)) \leq E(x)E(y).$$

Consequently,

$$\max(P_{\text{plus}}(x), P_{\text{plus}}(y)) + \min(E(x), E(y)) \leq E(x)E(y).$$

Sharper local form

Choose controlling prime powers $p^a \parallel x$ and $q^b \parallel y$ such that $S(p^a)=S(q^b)=m$. The primes are distinct. If $p < q$ and $m=pqc$, then

$$a \geq qc + 1, \quad b \geq pc.$$

If the smaller controlling prime belongs to y , interchange x and y . No uniqueness assumption on a controlling prime power is needed.

Equivalent fiber viewpoint

A fiber of S is a set $S^{-1}(m)$. The theorem says that any two coprime members of the same nontrivial fiber must carry large, cross-coupled prime exponents. The extra $+\min$ term is essential and sharp.

2. Proof of the coprime-fiber theorem

Step 1: reduce to activating prime powers

If z has prime-power factorization $z = \text{product } l^e$, then

$$S(z) = \max \{ S(l^e) : l^e \parallel z \}.$$

Hence there exist $p^a \parallel x$ and $q^b \parallel y$ with $S(p^a)=S(q^b)=m$. Minimality of m gives

$$v_p((m-1)!) < a \leq v_p(m!), \quad v_q((m-1)!) < b \leq v_q(m!).$$

Step 2: both activating primes divide the common value

If p did not divide m , multiplying $(m-1)!$ by m would not change its p -adic valuation, contradicting the displayed strict-to-weak jump. Thus p divides m ; similarly q divides m . Because x and y are coprime while p divides x and q divides y , p and q are distinct. Therefore pq divides m .

Step 3: Legendre's formula gives cross-exponent bounds

Assume $p < q$ and write $m = pqc$ with $c \geq 1$. Then

$$\begin{aligned} v_p((m-1)!) & \geq \text{floor}((m-1)/p) + \text{floor}((m-1)/p^2) \\ & \geq (qc-1) + 1 = qc. \end{aligned}$$

The second floor contributes at least one because $p^2 < m$: indeed $p < q$ and $c \geq 1$. Since a is strictly larger than $v_p((m-1)!)$, this yields $a \geq qc+1$. Likewise,

$$v_q((m-1)!) \geq \text{floor}((m-1)/q) = pc-1, \text{ so } b \geq pc.$$

Step 4: multiply the cross-bounds

Put $A = E(x)$ and $B = E(y)$. In this orientation $A \geq qc+1$ and $B \geq pc$. Therefore

$$(A-1)B \geq (qc)(pc) = pqc^2 = mc \geq m.$$

Thus $m+B \leq AB$. Since $\min(A,B) \leq B$, the first inequality of Theorem 1 follows. The other orientation is symmetric. Finally, x and y both divide $m!$, so each of their prime factors is at most m . Replacing m on the left by any such prime factor proves the Pplus consequence.

3. Sharpness and equality

Let $p < q < 2p$ be primes. Legendre's formula gives

$$v_p((pq-1)!) = q, \quad v_q((pq-1)!) = p-1.$$

Multiplication by pq adds one copy of both primes. It follows that

$$S(p^{q+1}) = S(q^p) = pq.$$

The two inputs are coprime, $E(p^{q+1})=q+1$, $E(q^p)=p$, and

$$pq + \min(q+1, p) = pq+p = (q+1)p.$$

So equality holds in Theorem 1. Bertrand's theorem supplies a prime q with $p < q < 2p$ for every prime p , giving infinitely many sharp pairs.

Equality classification at controller level

Tracing equality through the proof shows that equality forces $c=1$, maximal exponents $q+1$ and p on the relevant sides, and $q < 2p$. The construction above proves these conditions sufficient. The smallest example is $x=16$, $y=9$, $m=6$.

4. Consequence for the Tuțescu conjecture

The conjecture asserts $S(n) \neq S(n+1)$ for every positive integer n . Consecutive integers are coprime. If a counterexample existed, Theorem 1 would imply

$$\begin{aligned} & \max(P_{\text{plus}}(n), P_{\text{plus}}(n+1)) \\ & \leq E(n)E(n+1) - \min(E(n), E(n+1)). \end{aligned}$$

The local bounds also imply that one neighbor is divisible by a fourth power and the other by a square. Hence the theorem already excludes every pair with one squarefree member and every pair in which both members are fourth-power-free.

Important status

This is a necessary condition for a counterexample, not a contradiction. For example, the consecutive pair 8,9 satisfies the factorization-only obstruction, but $S(8)=4$ and $S(9)=6$.

5. Quantitative smooth-number reduction

Let $C(X)$ count hypothetical Tuţescu counterexamples $n \leq X$. Since $E(n), E(n+1) \leq \log_2(X+1)$, the prime-factor inequality makes both n and $n+1$ Y -smooth with

$$Y = (\log_2(X+1))^2.$$

Consequently,

$$C(X) \leq \Psi(X, Y) = X^{(1/2+o(1))}.$$

Here $\Psi(X, Y)$ counts integers at most X whose prime factors are all at most Y . The final equality is the standard estimate $\Psi(X, (\log X)^\alpha) = X^{(1-1/\alpha+o(1))}$ with $\alpha=2$. This is a power-saving structural reduction.

Comparison with the normal-order route

The classical fact $S(n) = P_{\text{plus}}(n)$ for almost all n only places counterexamples in a density-zero exceptional set. Ivić's sharper estimate for that exception set is of order

$$X \exp(-(\sqrt{2}+o(1)) \sqrt{\log X \log \log X}) = X^{(1-o(1))}.$$

The new coprime-fiber obstruction instead gives $X^{(1/2+o(1))}$.

6. Computational audit

The proof is independent of computation. Two reproducible checks were run:

1. The activation audit enumerated 5,004,826 pairs of activating prime-power configurations with common value $m \leq 10^6$. It found 0 violations, 8,097 sharp configurations, and 0 sharpness-classification mismatches.
2. The consecutive sieve checked $n \leq 10^8$. It found 0 equalities $S(n) = S(n+1)$. Only 30 pairs survived the new factorization-only necessary condition, and direct Kempner values eliminated all 30.

Exact reproduction commands

```
lake build
python3 scripts/audit_activations.py --max-value 1000000
g++ -O3 -std=c++20 scripts/search_consecutive.cpp -o /tmp/search_consecutive
/tmp/search_consecutive --limit 100000000
```

7. Lean/Mathlib verification

The formalization uses an actual graph predicate `IsKempnerValue x m`, not an algebraic surrogate. It proves activation existence from a fiber witness, valuation-jump lemmas, cross-exponent bounds, and the global coprime theorem.

Main checked declarations

```
coprime_activated_fiber_bound
coprime_kempner_fiber_bound
coprime_kempner_fiber_bound_sub
prime_factor_add_min_le_exponent_product
```

Source: `KempnerResearch/Basic.lean`. The main theorem has hypotheses $x, y > 1$, `IsKempnerValue x m`, `IsKempnerValue y m`, and `Nat.Coprime x y`, and concludes

$$m + \min(\text{maxPrimeExponent } x) (\text{maxPrimeExponent } y) \leq \text{maxPrimeExponent } x * \text{maxPrimeExponent } y.$$

The project builds with Lean v4.34.0-rc1 and the Mathlib revision pinned in `lake-manifest.json`. From the repository root, the checked command is:

```
lake build
```

8. Falsified stronger variants

- Strict inequality is false: $x=16$, $y=9$, $m=6$ gives equality.
- Replacing $E(x)E(y)$ by $(E(x)-1)(E(y)-1)$ is false for the same pair.
- Replacing the local lower bound $qc+1$ by $qc+2$ is false for $p=2$, $q=3$, $c=1$.
- The smoothness obstruction alone does not settle adjacency: 8,9 survives it.

Interpretation

These failures are useful boundary data. They show that the $+\min$ term, the extra single p -adic copy, and the distinction between a necessary structural filter and a proof of the conjecture are all genuine.

9. Expanded novelty audit

zbMATH Open (Zentralblatt): direct database coverage

- 257 records contain the exact phrase 'Smarandache function' in an indexed field; 155 match in titles and 174 in reviews or summaries.
- Seven records match Tutescu or Tuțescu; the number-theoretic conjecture paper is MR1650388 / Zbl 1008.11508.
- All 12 review/summary records containing both 'Smarandache function' and ' $S(n+1)$ ' were screened.
- Review/summary queries also combined the function name with Legendre, valuation, p-adic, exponent, prime power, same value, equal values, fiber, preimage, inverse image, smooth, coprime, maximal exponent, and maximum exponent. Plausible hits were inspected.
- No indexed review or summary states the coprime-fiber inequality, its local $qc+1/pc$ bounds, its sharp family, or the $X^{(1/2+o(1))}$ consequence.
- The indexed reference-text search found no citation under Prodanescu, Tutescu, or the exact conjecture-paper title; that index has incomplete reference coverage.

MathSciNet: public coverage and limitation

Full MathSciNet Anywhere, Review Text, and References searching redirected to institutional authentication in this environment. The public MR Lookup service was searched instead. Because it returns at most three regular results per query, 'Smarandache function' was queried separately for every year 1980-2026 and supplemented with targeted title conjunctions.

- The year-sliced public search produced 51 distinct MR records.
- There were no title hits coupling the function with consecutive, equal, equality, fiber, inverse, or exponent.
- The three coprime title hits concern a distinct 0-1 characteristic function, not coprime inputs sharing a Kempner value.
- Exact searches recovered the main known records and authors, including MR1650388, MR1364859, MR1416986, MR1294796, and the three editions/versions MR1294791, MR1361855, MR1398974.

Nearest prior results found

Ashbacher observes that if equal values are written $kp=rq$ using activating primes from the two coprime inputs, then q divides k and p divides r . Prodanescu and Tutescu prove the weaker fact that an input and its Kempner value share a prime factor. Neither source obtains the extra Legendre copy for the smaller prime or a global exponent-product bound.

Novelty conclusion

The theorem is proved and the expanded audit supports 'apparently novel.' It is still not 'certified novel': the MathSciNet search was not subscriber-level full text, database indexing can be incomplete, terminology varies, and an expert referee could locate an equivalent result.

10. Independent-review checklist

A second model or human referee should try to break the result at these points:

1. Verify the max-over-prime-powers identity supplies at least one activating component on each side without assuming uniqueness.
2. Check that activation at m forces the activating prime to divide m , and that coprimality forces the two activating primes to differ.
3. Check the extra floor term $\text{floor}((m-1)/p^2) \geq 1$ when $p < q$ and $m = pq$; this is the source of $qc+1$ rather than qc .
4. Trace every equality condition in $(A-1)B \geq mc \geq m$ and verify the $p < q < 2p$ construction.
5. Check that the P plus bound follows because x and y divide $m!$, then verify the smooth-number counting exponent $1/2$.
6. Run a subscriber-level MathSciNet Anywhere/Review Text/References search and search non-indexed Smarandache Function Journal archives for differently phrased equivalents.

References and stable identifiers

- [1] I. Prodanescu and L. Tutescu, *On a conjecture concerning the Smarandache function*, Smarandache Notions J. 9 (1998), 104-105. MR1650388; Zbl 1008.11508. [Full text](#).
- [2] C. Ashbacher, *An introduction to the Smarandache function*, Erhus University Press, 1995. MR1364859; Zbl 0834.11002. [Full text](#).
- [3] A. A. Mullin, *On the Smarandache function and the fixed-point theory of numbers*, Smarandache Notions J. 7 (1996), 107. MR1416986; Zbl 0885.11012.
- [4] A. Ivić, *On a problem of Erdős involving the largest prime factor of n* , Monatsh. Math. 145 (2005), 35-46. Zbl 1098.11046. [arXiv](#).
- [5] A. Hildebrand and G. Tenenbaum, *On integers free of large prime factors*, Trans. Amer. Math. Soc. 296 (1986), 265-290. [Related survey source](#).
- [6] [zbMATH Open](#) and [MathSciNet MR Lookup](#), database audit performed 2026-08-20.

Machine-readable bottom line

PROVED: for coprime $x, y > 1$ with $S(x) = S(y) = m$, $m + \min(E(x), E(y)) \leq E(x)E(y)$, sharply.
 CONSEQUENCE: hypothetical $S(n) = S(n+1)$ values up to X form a set of size at most $X^{1/2+o(1)}$. OPEN: existence is not ruled out. NOVELTY: apparently new after the stated audit, not historically certified.